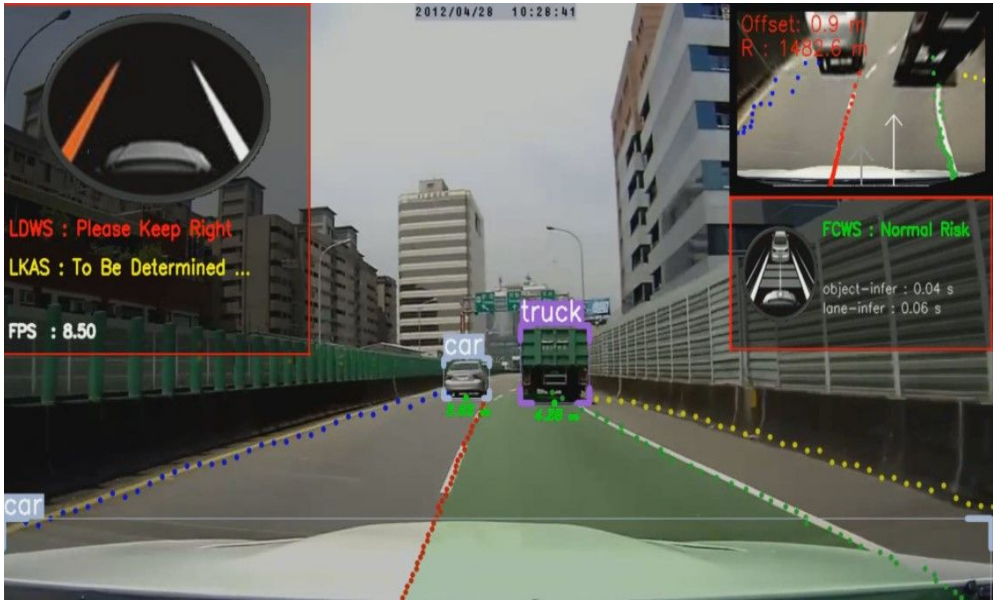
Advanced Driver Assistance for Teleoperation use cases



ADAS Features Highlighted

Core ADAS Functions

- **FCWS (Front Collision Warning System)** → alerts to imminent frontal collisions.
- **LDWS (Lane Departure Warning System)** → warns on unintended lane departure.
- **LKAS (Lane Keeping Assist System)** → provides corrective steering guidance.

My Implementations

- **Object Detection** → YOLOv8m, EfficientDet (ONNX/TensorRT for acceleration).
- **Object Tracking** → ByteTrack for stable trajectory tracking of vehicles.
- **Lane Detection** → Ultra-Fast-Lane-Detection v2 (UFLD-v2) model (Tusimple with resnet18).

Key Point

- All modules are **vision-based only** — no radar or lidar — making the system **lightweight and cost-effective**.

Motivation (Teleoperation Focus)

Benefits of ADAS in Teleoperation

Improved Situational Awareness

- FCWS warns of frontal obstacles the operator may miss on limited video feed.
- LDWS highlights unintended lane drift, enhancing lane awareness.

Reduced Operator Workload

- LKAS provides steering guidance, lowering cognitive demand.
- Automatic alerts let operator focus on higher-level navigation.

Compensation for Latency

- Predictive warnings mitigate delayed human reaction due to network lag.
- Early feedback ensures safer remote maneuvering.

Enhanced Safety & Reliability

- Continuous monitoring reduces risk of accidents in remote driving.
- Builds operator confidence in teleoperation systems.

How the System Works

1. Input Capture

- Rover/vehicle camera provides real-time video stream.

2. Perception Modules

- **Object Detection (YOLO)** → detect vehicles, pedestrians, obstacles.
- **Object Tracking (ByteTrack)** → track movement, estimate trajectories.
- **Lane Detection (UFLD-v2)** → detect lane boundaries and centerline.

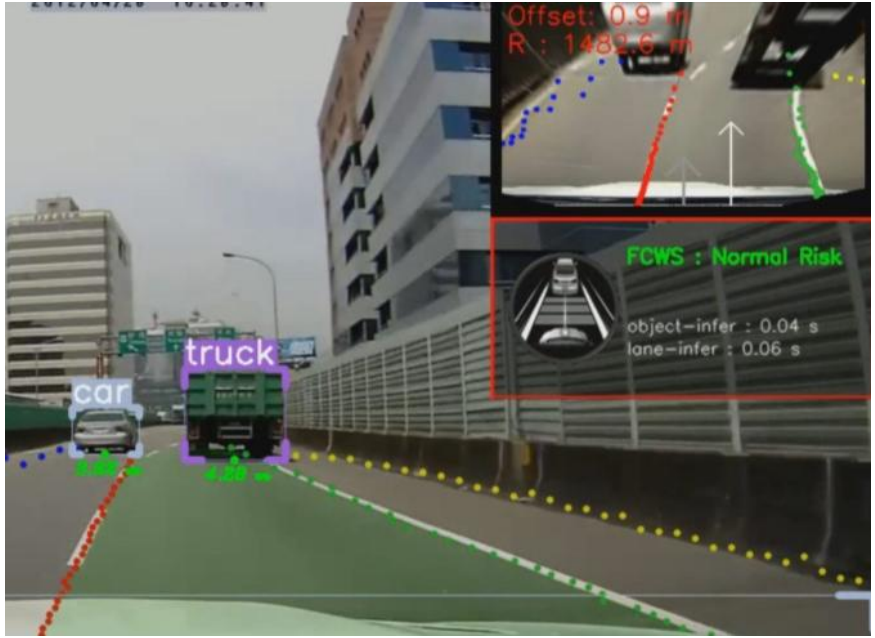
3. Decision Logic

- **FCWS** → compute time-to-collision, trigger warning if threshold crossed.
- **LDWS** → monitor lateral offset, warn on unintended lane departure.
- **LKAS** → calculate corrective steering angle from lane curvature.

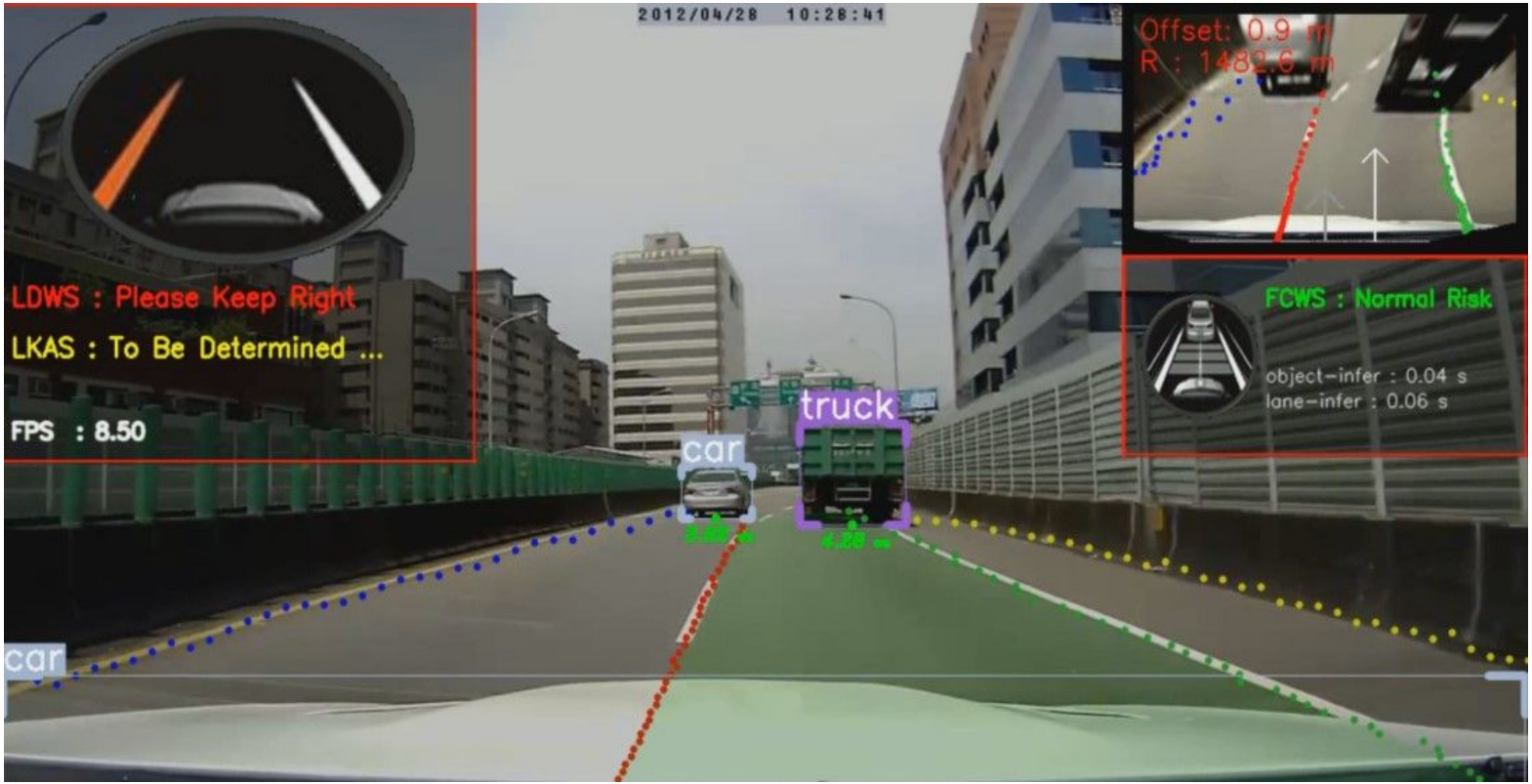
4. Output / Feedback (need to work on)

- **To Operator** → visual/audio alerts, steering guidance overlay.
- **To Rover (Using ROS 2)** → closed-loop feedback for steering & velocity control.

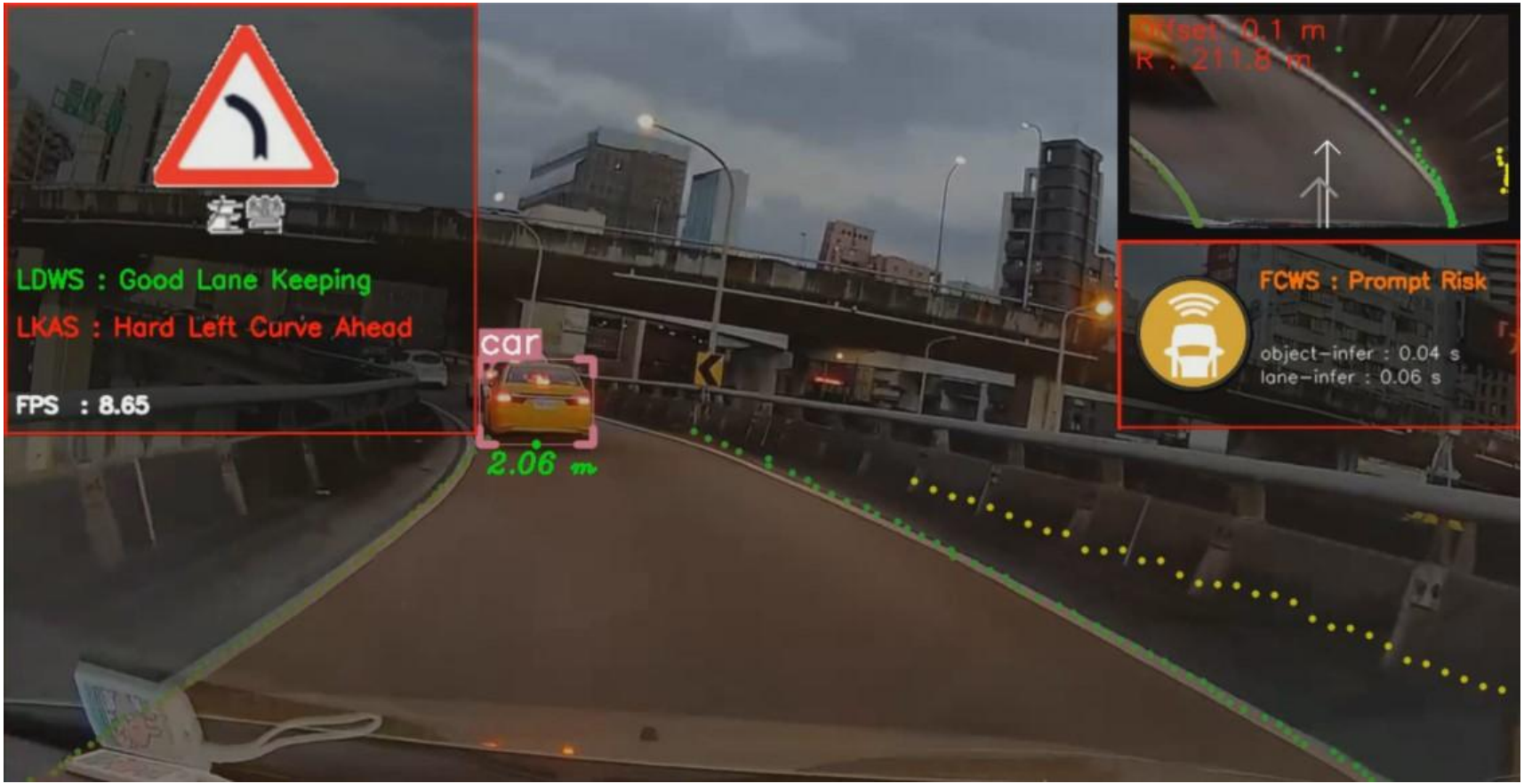
Bird’s eye view (BEV) Use



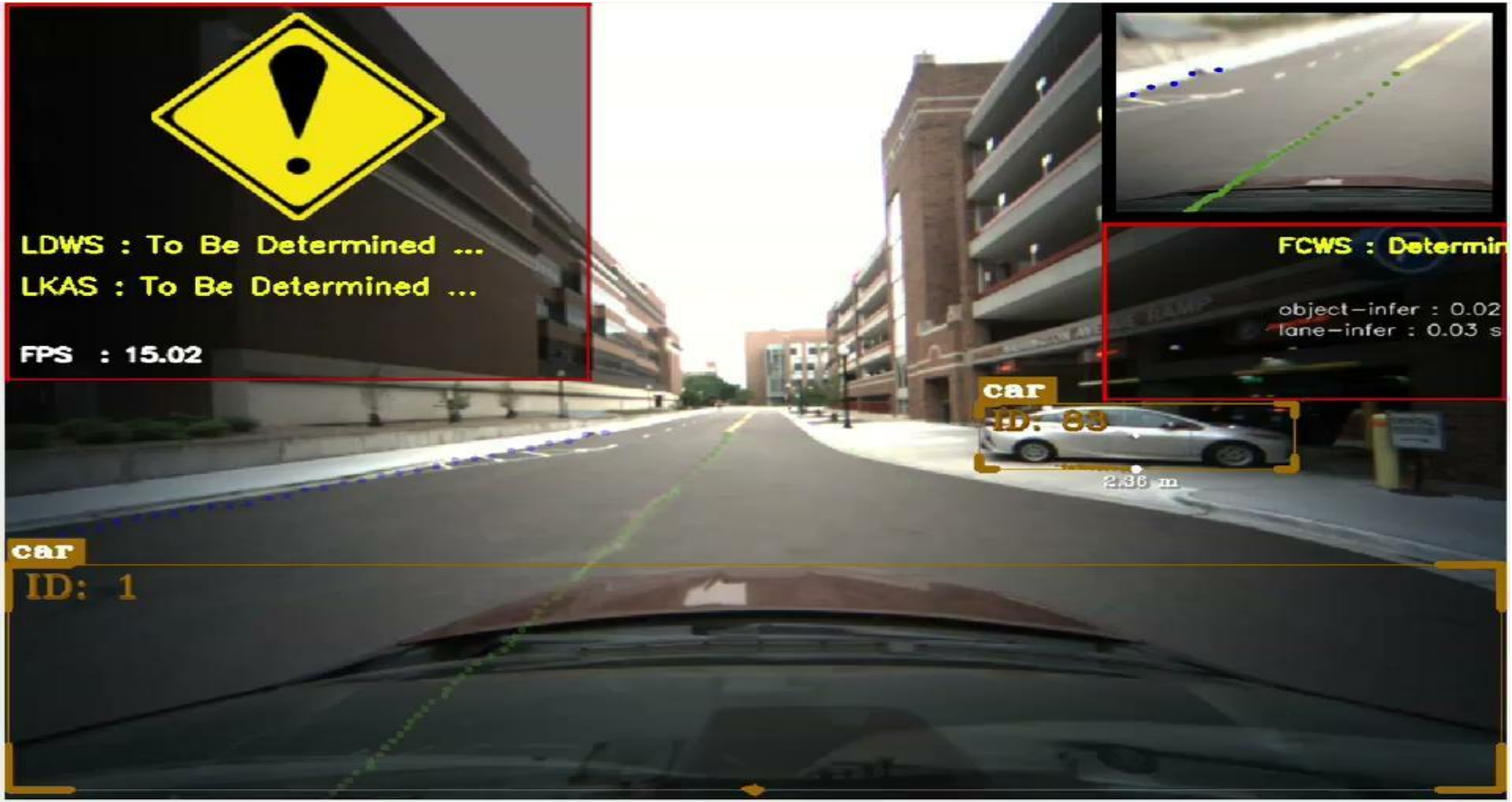
Lane Departure Warning System (LDWS)



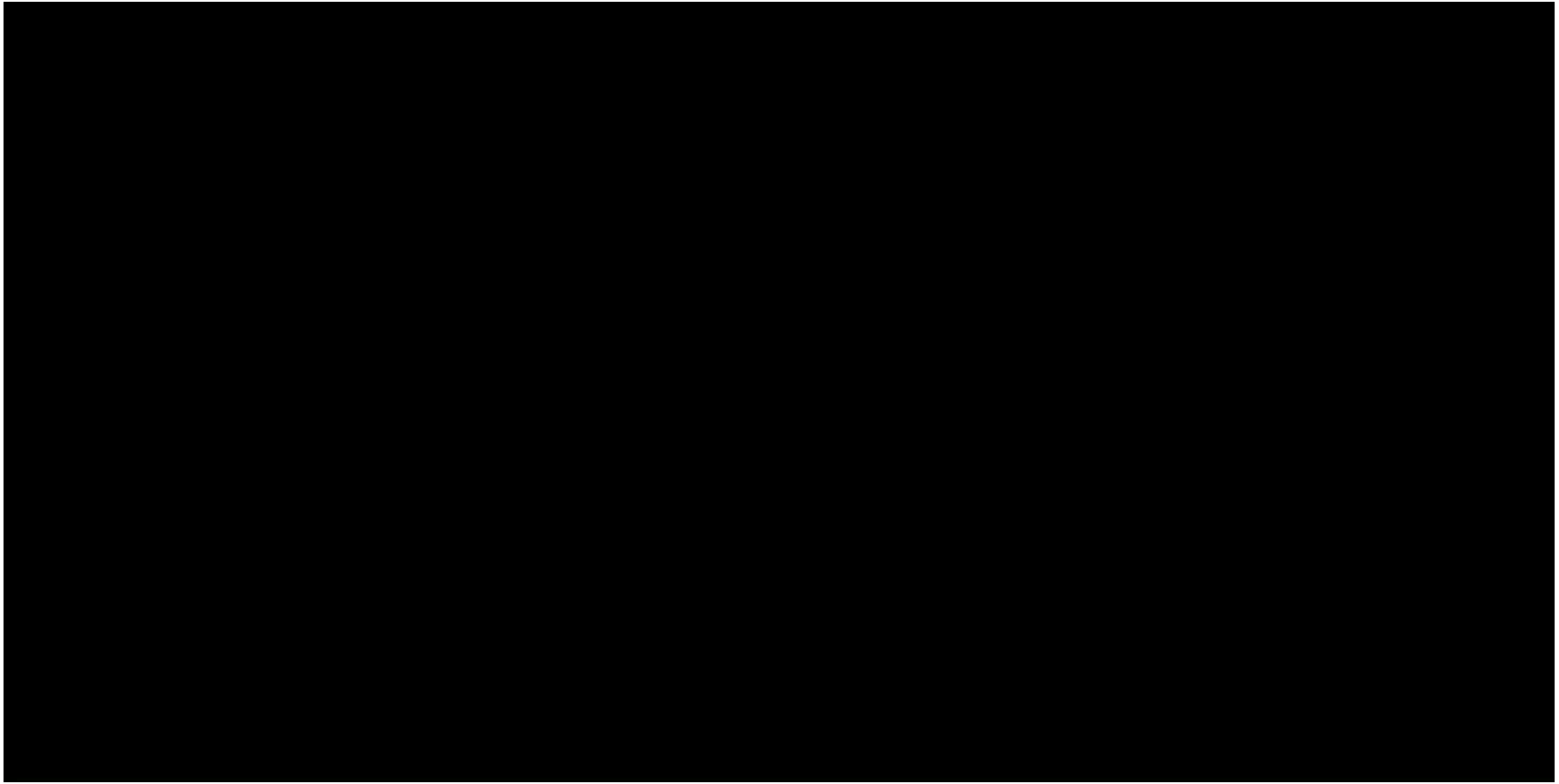
Lane Keeping Assist System (LKAS)



Results on MnCAV Vehicle



ADAS + BEV for Teleoperator



Future Work:

Integration with our Vehicle - DONE

Deploy ADAS perception modules for MnCAV using **ROS 2** nodes.

Closed-loop Control - DONE

Connect FCWS, LDWS, LKAS outputs to rover's **steering and velocity controllers** and implement real-time corrective feedback.

Latency Compensation- Working

Provide predictive warnings and steering guidance to counter network delays during teleoperation.

Next - Integration of VLM(Vision language Model) to ADAS